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Track Selected: Track 1 - RAG

GitHub Repository URL: <https://github.com/akwasianing/clinical-decision-support-rag-diabetes>

SECTION 1: PROJECT TITLE & ONE-SENTENCE DESCRIPTION

Project Title: Clinical Decision Support Assistant Using RAG for Type 2 Diabetes Management

Description: A Retrieval-Augmented Generation (RAG) clinical decision support system that assists healthcare professionals, researchers, and students in answering Type 2 Diabetes management questions using evidence-based clinical guidelines and providing citation-supported responses.

SECTION 2: PROBLEM & MOTIVATION

Healthcare professionals, students, and researchers often spend significant time searching through lengthy clinical guidelines to find evidence-based recommendations for Type 2 Diabetes management. This project aims to make that process easier by developing a Retrieval-Augmented Generation (RAG) assistant that retrieves relevant guideline sections and generates concise, citation-supported responses. I am particularly interested in this project because it combines healthcare and generative AI, while providing an opportunity to apply large language models to a real-world clinical decision support problem.

SECTION 3: DATA OR API ACCESS

Data source:

American Diabetes Association (ADA) Standards of Care in Diabetes - 2026 (published in the Diabetes Care journal supplement).

Access status: Already have it

If you need to obtain it, how?

Not applicable. The ADA Standards of Care in Diabetes - 2026 guideline documents have already been downloaded and will be used as the primary document corpus for the RAG system.

SECTION 4: TECHNICAL APPROACH

The project will use the American Diabetes Association (ADA) Standards of Care in Diabetes - 2026 guideline documents as the primary knowledge base for a Retrieval-Augmented Generation (RAG) system. The documents will be processed, divided into smaller text chunks, converted into embeddings, and stored in a vector database for efficient retrieval. When a user asks a Type 2 Diabetes management question, the system will retrieve the most relevant guideline sections and use them to generate an evidence-based response with citations. The project will be

developed using Python and appropriate tools for document processing, embedding generation, vector storage, retrieval, and user interaction. Likely tools include LangChain, ChromaDB among others.

SECTION 5: SCOPE - WHAT YOU WILL DEFINITELY BUILD

1. A document ingestion pipeline that loads and processes the ADA Standards of Care in Diabetes—2026 guideline documents.
2. A vector database that stores embeddings of the guideline content for semantic retrieval.
3. A question-answering system that retrieves relevant guideline sections in response to Type 2 Diabetes management questions.
4. Citation-supported responses that reference the source guideline documents used to generate answers.
5. A simple interface for entering questions and displaying answers.

SECTION 6: SCOPE - WHAT YOU WILL NOT BUILD

This project will not include integration with Electronic Health Record (EHR) systems or the use of real patient data. The system will not provide patient-specific diagnoses, treatment recommendations, or medical advice, and its scope will be limited to information contained within the ADA Standards of Care in Diabetes—2026 guidelines. In addition, the project will not involve fine-tuning or training a custom large language model, and it will not be developed as a production-ready or HIPAA-compliant clinical decision support system.

SECTION 7: WEEK-BY-WEEK ROUGH PLAN

Weeks 1–2:

- Review RAG concepts and the ADA Standards of Care in Diabetes—2026 guidelines.
- Organize the guideline documents and set up the Python environment.

Week 3: Build baseline RAG system

- Load and process the guideline documents.
- Generate embeddings and store them in a vector database.
- Test basic retrieval using sample diabetes-related questions.

Week 4: Improve retrieval and answer generation

- Integrate a large language model with the retrieval pipeline.
- Generate citation-supported responses using retrieved guideline sections.

Week 5: Add additional functionality

- Improve retrieval performance and response quality.
- Implement a fallback response when relevant information cannot be found.

Week 6: Evaluation

- Create a set of Type 2 Diabetes management questions for testing.
- Evaluate retrieval relevance, answer quality, and citation accuracy.

Weeks 7–8: Polish, report, and demo

- Refine the system and conduct final testing.
- Prepare project documentation, final report, and presentation.
- Demonstrate the completed system.

SECTION 8: BIGGEST RISKS

Risk 1:

The retrieval system may not always return the most relevant sections of the ADA guidelines, which could affect the quality and accuracy of the responses.

Backup plan:

Different chunk sizes and retrieval settings will be tested to improve performance. If needed, a smaller subset of the ADA guidelines will be used to improve retrieval accuracy and simplify testing.

Risk 2:

The generated responses may occasionally include information that is not fully supported by the retrieved guideline content.

Backup plan:

Responses will be based on retrieved guideline sections and accompanied by source citations. A fallback response will also be implemented when sufficient supporting information cannot be found in the document corpus.